PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION

FORM THREE TERMINAL EXAMINATION

CODE:033 **BIOLOGY**

TIME: 3: 00 HRS **29th May, 2026**

INSTRUCTIONS

1. This paper consists of three sections A, B and C, with a total of 11 questions.
2. Answer all question in section A and B, and two (2) from section C.
3. Except diagrams that must be drawn by pencil all writing should be in blue or black ink.
4. Cellular phone and any unauthorised materials are not allowed in the examination room.
5. Write your name on every page of your answer sheet provided.

SECTION A (16 Marks)

Answer **all** the questions

1. For each of the following questions (i) to (x) choose the correct answer from among the given alternatives and write its letter in the provided answering space.

(i) A form three student went to hospital laboratory for the check-up whereby he was diagnosed with the sleeping sickness. Suggest the possible vector of such disease.

A. Butterfly
B. Housefly
C. Tsetse fly
D. Mosquitoes

(ii) Assume you have been assigned by school manager to cross-section cutting down the mango stem. Recommend on the arrangement vascular bundle.

A. Ring pattern
B. Random pattern
C. Peripheral pattern
D. Scattered pattern

(iii) What is the main product of anaerobic respiration in animals?

A. Alcohol
B. Lactic acid
C. Uric acid
D. Pyruvic acid

(iv) Mr. John wants export glass materials from Dar es Salam to Arusha but he fears on its safety of his language. Recommend the biological term that he could put on item box as a safety sign

A. Biohazard
B. Fragile
C. Toxic
D. Flammable

(v) Dr Maryam emphasize women to breast feed their babies to strengthen their body immunity. The aim of Dr Maryam advice is to help babies to acquire:

A. Natural passive immunity
B. Natural active immunity
C. Artificial passive immunity
D. Artificial active immunity

(vi) Which of the following is the excretory organ in human?

A. Mouth
B. Kidney
C. Pancreas
D. Stomach

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(vii) The phylum Chordata include all animals that have

A. Warm dry skin
B. Post anal tail
C. Mammary gland
D. A backbone

(viii) Which part is the functional unit of the kidney

A. Pelvis
B. Nephron
C. Medulla
D. Cortex

(ix)The green organisms are known as producers because they

A. They make food from simple substance
B. Produce chlorophyll in chloroplasts
C. Produce green leaves
D. Contain fruits and seeds

(x) The joint in human body which allows movement in all direction is known as

A. Pivot
B. Hinge
C. Ball and socket
D. Peg and socket

2. Match the hormones in list A with their corresponding effects in human body in list B by writing the letter of the correct response beside the item number in the answer sheet provided.

LIST A	LIST B
(i) Oxytocin (ii) Vasopressin (iii)Prolactin (iv)Luteinizing (v) Calcitonin (vi)Progesterone	A. Acts on kidney tubeless to regulate amount of water to be reabsorbed B. Stimulate production of milk to the lactating mother C. Causes the release of ovum from the ovary D. Causes the development of graafian follicle in ovary E. It is produced during pregnancy to prepare the body to act or run F. It converts glycogen to glucose in liver cells G. Secreted by thyroid glands to regulate the level of calcium in blood H. It stimulates the contraction of uterus during child birth I. It facilitates the uptake of glucose in cells, muscles and liver J. Maintain pregnancy by preventing production of ovum and contraction of the uterine wall.

SECTION B

3. (a) Plants are able to make their own food through photosynthesis, still the animal benefit much from this process. As a form three student explain any four (4) benefits of such process to animals.
(b) The small intestine is an important part of alimentary canal that makes maximum absorption of several nutrients, explain five (5) features that make it efficient in performing its role.

4. During a laboratory experiment red blood cells and cucumber cells were each placed in a distilled water, student observed that the two (2) types of cells reacted differently.
(a) What do you think will happen to:
(i) The red blood cell
(ii) The cucumber cell
(b) Why do you think the two (2) types of cell react differently when placed in distilled water?
(c) Name and explain the biological process responsible for these changes.

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5. (a) It is possible to breath by using both mouth and nose simultaneously or one of them at a time, but one passage is the best.
   (i) Which passage is the best for breathing?
   (ii) Give reasons for your answers.
   (b) Differentiate between breathing and gaseous exchange. Give any three (3) points.
   (c) Describe the essential adaptive features of respiratory surface. Give three (3) points

6. (a) (i). Define the term Taxonomic hierarchy.
   (ii). Using man as an example illustrate the concept of Taxonomic hierarchy.
   (b) Explain any four (4) advantages of using artificial system of classification.

7. (a) Moringe on his way to school came across a rattlesnake on the path coming right toward him, in no time he found himself up a tree.
   (i). What gave him the ability to climb the tree so quick?
   (ii). Identify the different processes that went on his body during this incidence.
   (b) Stunted growth and severe mental retardation during early stage of development may be due to the under-secretion of a certain chemical substance
   (i) Identify this chemical substance
   (ii) What name is given to the condition described in (b)?

8. (a) Explain three (3) types of muscles found in mammals
   (b) Briefly explain how muscles are adapted to their role. Give any four (4) points.

SECTION C: (30 Marks)

9. Life on the earth would have been impossible without the presence of bacteria. Justify this statement by giving six (6) points.

10. With the aid of a well labelled diagram, describe the internal parts of the mammalian heart.

11. Some chemical reactions take place inside the human body and produce substances that can be harmful if they are allowed to accumulate, these are known as excretory products. Based on excretory organs involved in each product, Explain the organ responsible for its excretion, where it is produced in the body, and the harmful effects it may have if it accumulates.

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